Version 7.4.6

PVsyst - Simulation report

Grid-Connected System

Project: Madrid_Fixed_Angle_PV_Syst

Variant: Madrid_150MW - Simulation for year no 25 - Simulation for year no 25

Unlimited sheds

System power: 132.7 MWp

Rosas - Spain

Author

University of Edinburgh (United kingdom)

Project summary

Geographical Site	Situation		Project settings	
Rosas	Latitude	40.42 °N	Albedo	0.20
Spain	Longitude	-3.59 °W		
	Altitude	664 m		
	Time zone	UTC+1		

Weather data
Rosas
PVGIS api TMY

System summary

Grid-Connected System
Simulation for year no 25

Unlimited sheds

PV Field Orientation
Sheds
Tilt 28 °
Azimuth 0 °

Near Shadings
Mutual shadings of sheds

User's needs
Unlimited load (grid)

System information

PV Array		Inverters	
Nb. of modules	228781 units	Nb. of units	17 units
Pnom total	132.7 MWp	Pnom total	116.9 MWac
		Pnom ratio	1.136

Results summary

Produced Energy	199633361 kWh/year	Specific production	1504 kWh/kWp/year	Perf. Ratio PR	71.58 %

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General parameters

Grid-Connected System — **Unlimited sheds**

PV Field Orientation

Orientation		Sheds configuration		Models used	
Sheds		Nb. of sheds	5 units	Transposition	Perez
Tilt	28 °	Unlimited sheds		Diffuse	Imported
Azimuth	0 °	**Sizes**		Circumsolar	separate
		Sheds spacing	6.60 m		
		Collector width	3.00 m		
		Ground Cov. Ratio (GCR)	45.5 %		
		Top inactive band	0.02 m		
		Bottom inactive band	0.02 m		
		Shading limit angle			
		Limit profile angle	19.8 °		

Horizon		Near Shadings	User's needs
Average Height	1.2 °	Mutual shadings of sheds	Unlimited load (grid)

PV Array Characteristics

PV module		Inverter	
Manufacturer	Generic	Manufacturer	Generic
Model	NB-JD580	Model	6250KVA-MV
(Original PVsyst database)		(Original PVsyst database)	
Unit Nom. Power	580 Wp	Unit Nom. Power	6874 kWac
Number of PV modules	228781 units	Number of inverters	17 units
Nominal (STC)	132.7 MWp	Total power	116858 kWac
Modules	9947 string x 23 In series	Operating voltage	875-1300 V
At operating cond. (50°C)		Max. power (=>25°C)	7186 kWac
Pmpp	122.9 MWp	Pnom ratio (DC:AC)	1.14
U mpp	923 V	Power sharing within this inverter	
I mpp	133056 A		

Total PV power		Total inverter power	
Nominal (STC)	132693 kWp	Total power	116858 kWac
Total	228781 modules	Max. power	122162 kWac
Module area	590999 m²	Number of inverters	17 units
Cell area	545626 m²	Pnom ratio	1.14

Array losses

Array Soiling Losses		Thermal Loss factor		DC wiring losses	
Loss Fraction	2.0 %	Module temperature according to irradiance		Global array res.	0.11 mΩ
		Uc (const)	29.0 W/m²K	Loss Fraction	1.5 % at STC
		Uv (wind)	0.0 W/m²K/m/s		

LID - Light Induced Degradation		Module Quality Loss		Module mismatch losses	
Loss Fraction	2.0 %	Loss Fraction	-1.3 %	Loss Fraction	2.0 % at MPP

Strings Mismatch loss		Module average degradation	
Loss Fraction	0.2 %	Year no	25
		Loss factor	0.4 %/year
		Mismatch due to degradation	
		Imp RMS dispersion	0.4 %/year
		Vmp RMS dispersion	0.4 %/year

Array losses

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

Spectral correction

FirstSolar model

Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0.85914	-0.02088	-0.0058853	0.12029	0.026814	-0.001781

Horizon definition

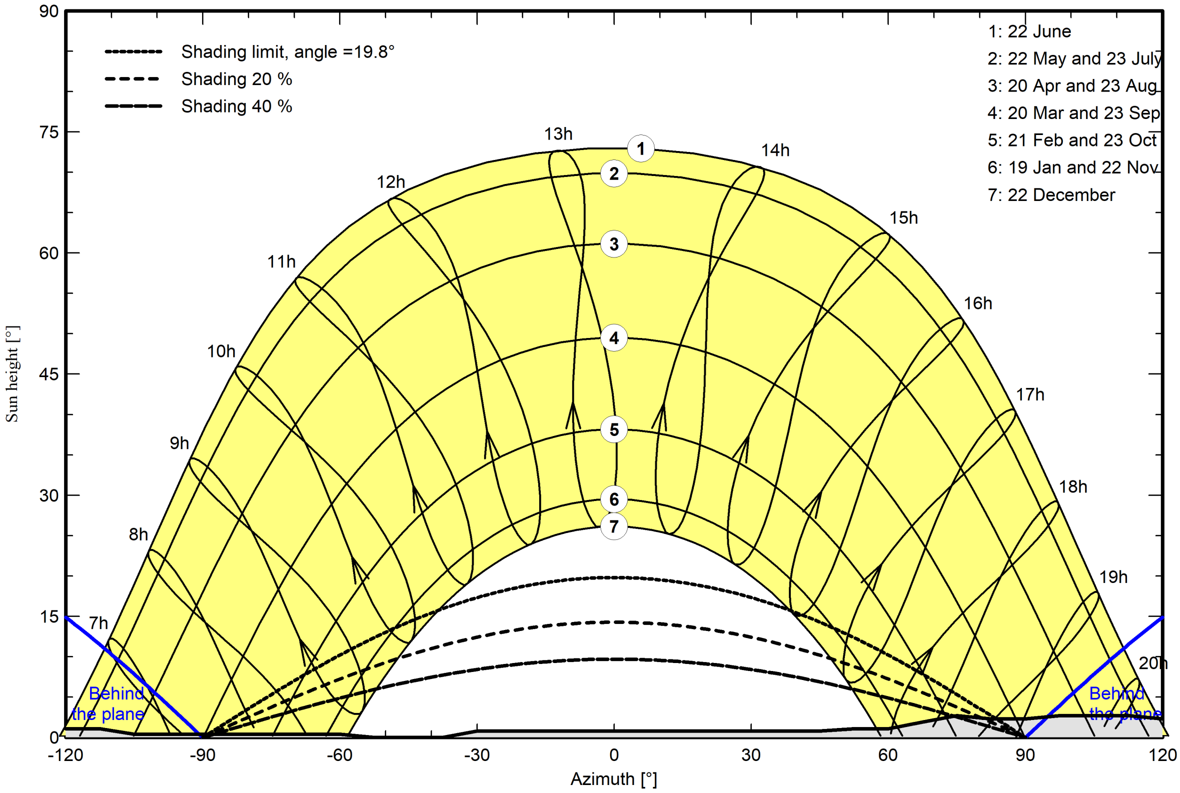
Horizon from PVGIS website API, Lat=40°25'13", Long=-3°35'27", Alt=664m

Average Height	1.2 °	Albedo Factor	0.96
Diffuse Factor	1.00	Albedo Fraction	100 %

Horizon profile

Azimuth [°]	-180	-173	-165	-143	-135	-113	-105	-60	-53	-38	-30	45	53	60
Height [°]	1.1	1.1	0.8	0.8	1.1	1.1	0.4	0.4	0.0	0.0	0.8	0.8	1.1	1.1
Azimuth [°]	68	75	83	90	98	113	120	128	135	158	165	173	180	
Height [°]	1.9	2.7	2.3	2.3	2.7	2.7	2.3	1.9	2.3	2.3	1.9	1.9	1.1	

Sun Paths (Height / Azimuth diagram)



Main results

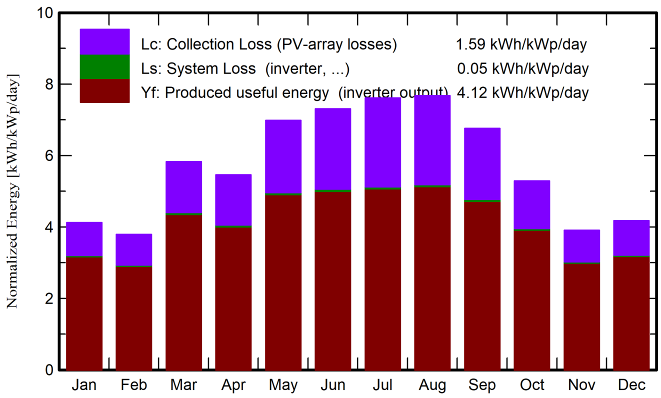
System Production

Produced Energy	199633361 kWh/year	Specific production	1504 kWh/kWp/year
		Perf. Ratio PR	71.58 %

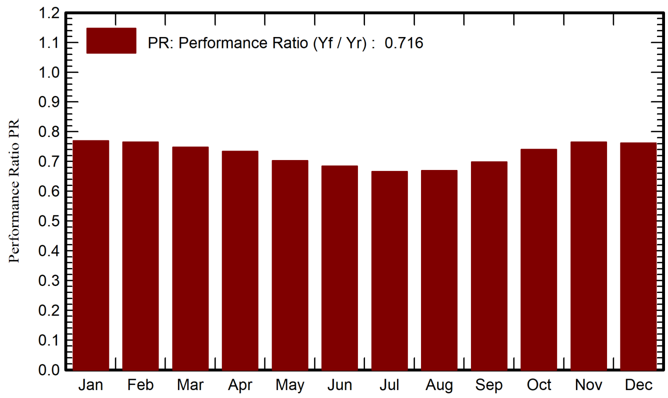
Economic evaluation

Investment		Yearly cost		LCOE	
Global	326,836,667.24 EUR	Annuities	7,141,381.18 EUR/yr	Energy cost	0.09 EUR/kWh
Specific	2.46 EUR/Wp	Run. costs	8,469,654.90 EUR/yr		
		Payback period	14.3 years		

Normalized productions (per installed kWp)



Performance Ratio PR



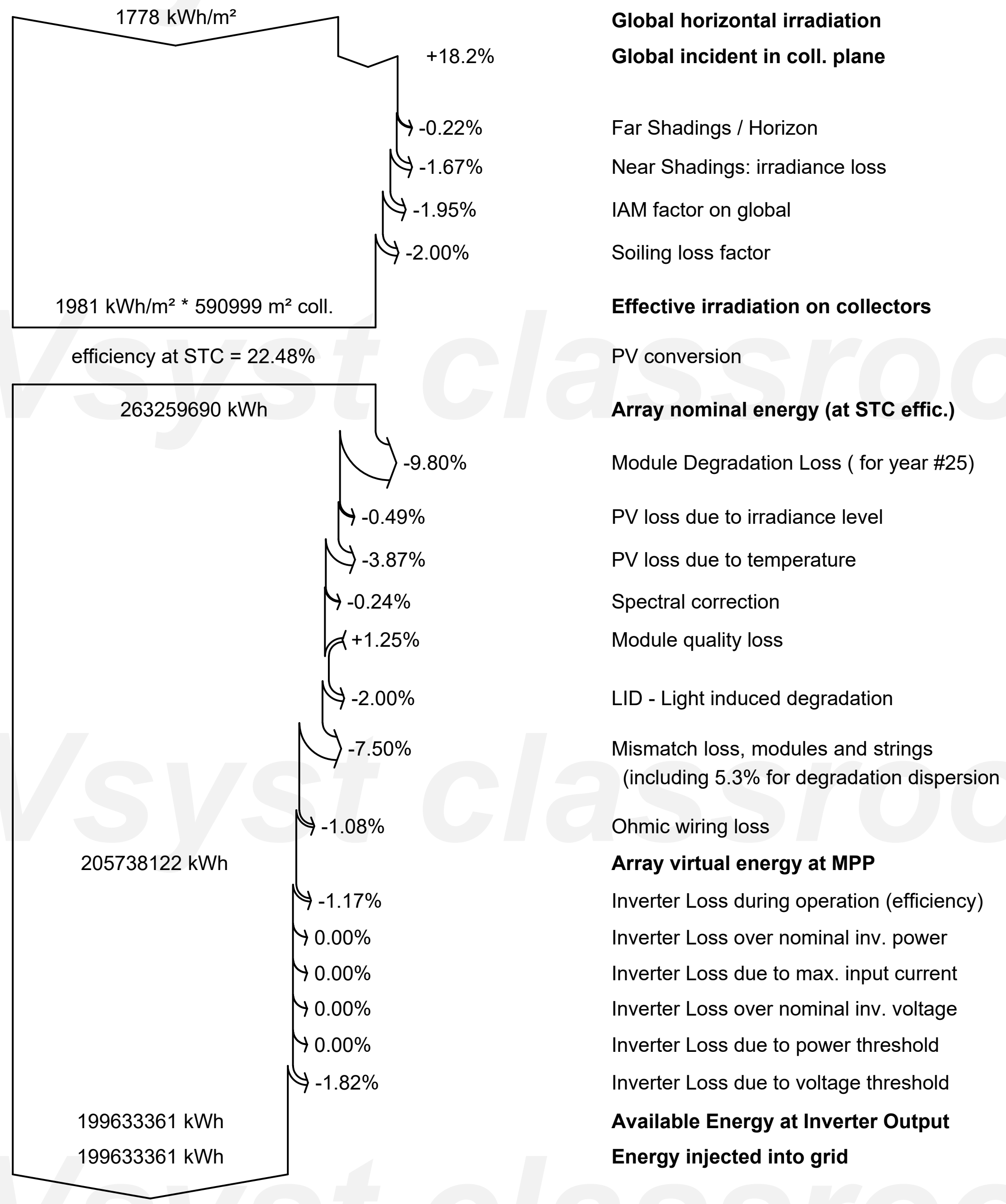
Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	75.2	23.27	5.77	127.8	119.7	13201861	13046545	0.769
February	77.1	37.12	6.28	106.1	99.6	10905750	10766676	0.765
March	142.0	44.83	9.58	180.6	171.3	18123580	17917451	0.748
April	151.7	67.03	11.94	163.8	154.3	16146518	15942180	0.733
May	215.3	63.56	19.22	216.6	204.4	20439224	20195464	0.703
June	226.9	66.81	23.62	219.3	206.7	20157629	19899833	0.684
July	239.5	62.66	27.81	236.1	222.9	21107168	20850826	0.666
August	221.9	50.72	27.04	237.8	225.4	21363925	21121010	0.669
September	166.8	44.76	22.92	202.9	192.8	19016788	18796572	0.698
October	117.7	42.39	16.64	164.0	155.2	16304060	16114835	0.741
November	74.0	27.90	7.67	117.3	110.1	12042035	11896740	0.764
December	69.5	19.76	5.63	129.5	119.1	13236876	13085230	0.762
Year	1777.8	550.82	15.40	2101.7	1981.5	202045414	199633361	0.716

Legends

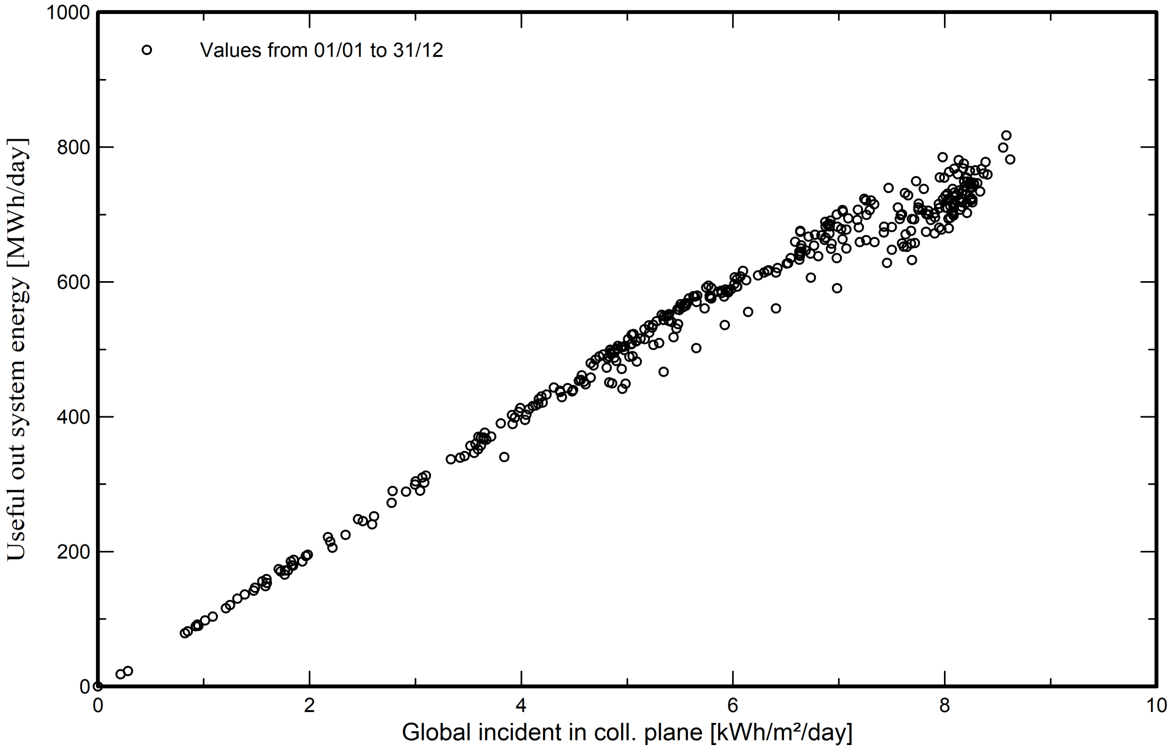
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram

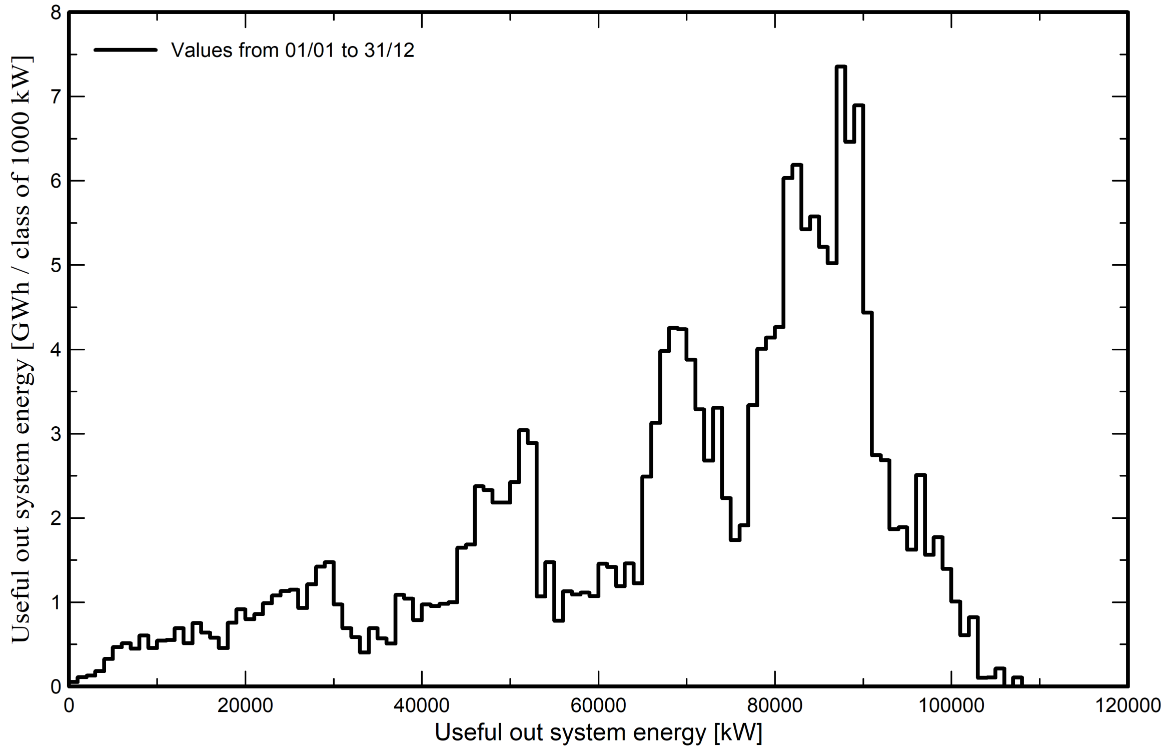


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



Aging Tool

Aging Parameters

Time span of simulation 25 years

Module average degradation

Loss factor 0.4 %/year

Mismatch due to degradation

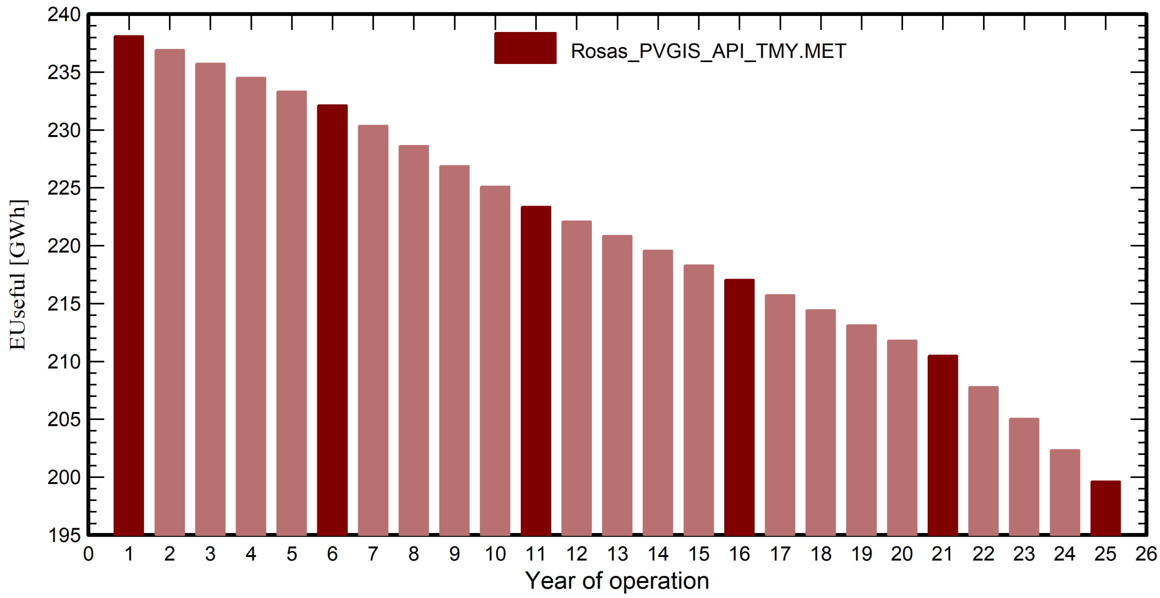
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

Weather data used in the simulation

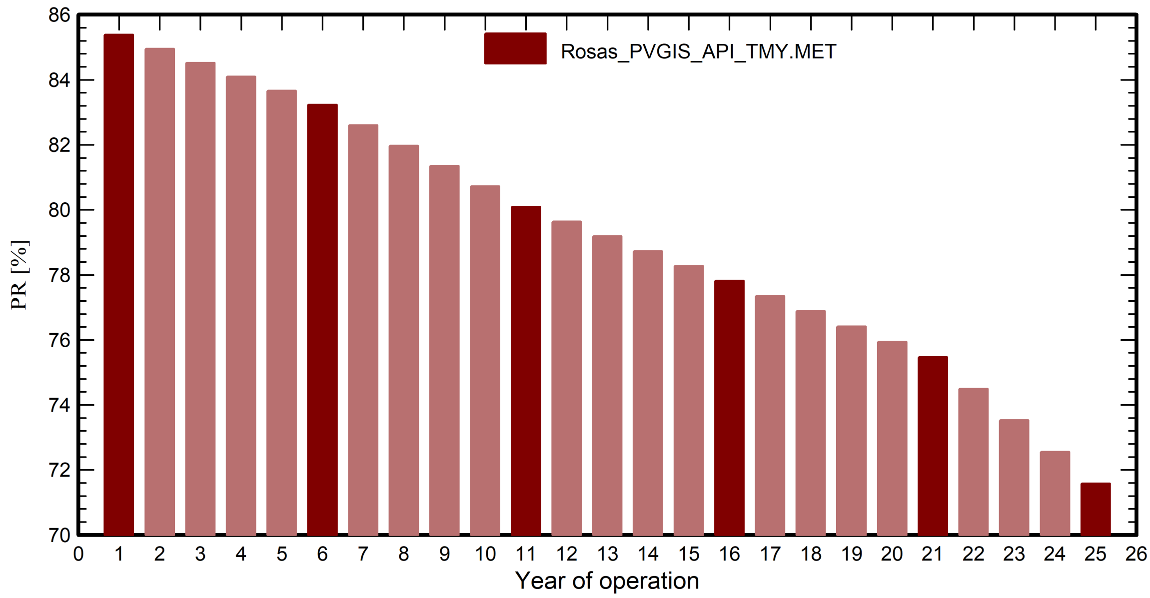
Rosas PVGIS API TMY

Years reference year

Useful out system energy



Performance Ratio



Aging Tool

Aging Parameters

Time span of simulation 25 years

Module average degradation

Loss factor 0.4 %/year

Mismatch due to degradation

Imp RMS dispersion 0.4 %/year

Vmp RMS dispersion 0.4 %/year

Weather data used in the simulation

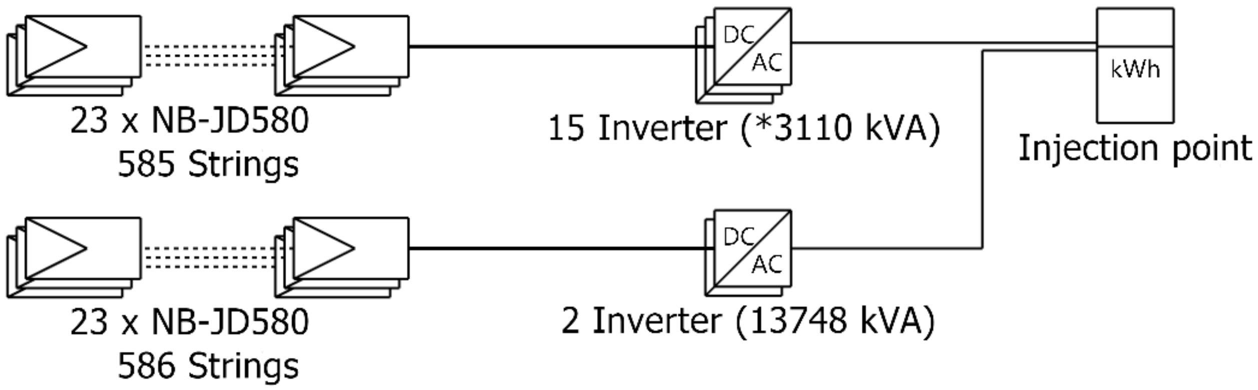
Rosas PVGIS API TMY

Years reference year

	EUseful	PR	PR loss
Year	GWh	%	%
1	238.1	85.38	-0.25
2	236.9	84.95	-0.75
3	235.7	84.52	-1.26
4	234.5	84.09	-1.76
5	233.3	83.66	-2.26
6	232.1	83.23	-2.76
7	230.4	82.60	-3.50
8	228.6	81.97	-4.23
9	226.9	81.35	-4.96
10	225.1	80.72	-5.70
11	223.4	80.09	-6.43
12	222.1	79.64	-6.96
13	220.8	79.18	-7.49
14	219.6	78.73	-8.02
15	218.3	78.27	-8.55
16	217.0	77.82	-9.08
17	215.7	77.35	-9.63
18	214.4	76.88	-10.18
19	213.1	76.41	-10.73
20	211.8	75.94	-11.28
21	210.5	75.47	-11.83
22	207.8	74.50	-12.97
23	205.0	73.52	-14.10
24	202.3	72.55	-15.24
25	199.6	71.58	-16.37

Single-line diagram

PVsyst V7.4.6
VCC, Simulation date:
28/03/24 18:02
with V7.4.6



PV module	NB-JD580
Inverter	6250KVA-MV
String	23 x NB-JD580

Madrid_Fixed_Angle_PV_Syst	University of Edinburgh (United kingd
VCC : Madrid_150MW - Simulation for year no 25 - Simulation for year no 25	28/03/24

Cost of the system

Installation costs

Item	Quantity units	Cost EUR	Total EUR
PV modules			
NB-JD580	228781	192.23	43,979,036.06
Supports for modules	228781	0.67	153,425.11
Inverters			
6250KVA-MV	17	55,884.65	950,039.12
Other components			
Wiring	150	65,375.50	9,806,325.08
Monitoring system, display screen	150	8,171.94	1,225,790.64
Measurement system, pyranometer	150	12,257.91	1,838,685.95
Studies and analysis			
Engineering	1	43,590.03	43,590.03
Installation			
Transport	1	9,016,500.06	9,016,500.06
Settings	150	776,334.07	116,450,110.32
Grid connection	150	61,289.53	9,193,429.76
Insurance			
Building insurance	1	204,298.44	204,298.44
Transport insurance	1	122,579.06	122,579.06
Land costs			
Land purchase	668083	154.30	103,085,206.90
Land preparation	150	65,375.50	9,806,325.08
Land taxes	1	0.00	11,494,000.57
Taxes			
VAT	1	0.00	9,467,325.06
		Total	326,836,667.24
		Depreciable asset	45,082,500.29

Operating costs

Item	Total EUR/year
Maintenance	
Provision for inverter replacement	38,001.56
Salaries	1,005,000.00
Repairs	210,000.00
Insurance	
Facilities insurance	300,000.00
Administrative, accounting	600,000.00
Total (OPEX)	2,153,001.56
Including inflation (10.00%)	8,469,654.90

System summary

Total installation cost	326,836,667.24 EUR
Operating costs (incl. inflation 10.00%/year)	8,469,654.90 EUR/year
Produced Energy	199633 MWh/year
Cost of produced energy (LCOE)	0.0923 EUR/kWh

Financial analysis

Simulation period

Project lifetime	25 years	Start year	2025

Income variation over time

Inflation	10.00 %/year
Production variation (aging)	Aging tool results
Discount rate	8.00 %/year

Income dependent expenses

Income tax rate	47.00 %/year
Other income tax	0.00 %/year
Dividends	0.00 %/year

Depreciable assets

Asset	Depreciation method	Depreciation period (years)	Salvage value (EUR)	Depreciable (EUR)
PV modules				
NB-JD580	Straight-line	20	0.00	43,979,036.06
Supports for modules	Straight-line	20	0.00	153,425.11
Inverters				
6250KVA-MV	Straight-line	20	0.00	950,039.12
		Total	0.00	45,082,500.29

Financing

Loan 1 - Interest-only bullet loan - 25 years	163,418,333.60 EUR	Interest rate: 0.01%/year
Loan 2 - Interest-only bullet loan - 25 years	163,418,333.60 EUR	Interest rate: 4.36%/year

Electricity sale

Feed-in tariff	0.43000 EUR/kWh
Duration of tariff warranty	20 years
Annual connection tax	0.00 EUR/kWh
Annual tariff variation	0.0 %/year
Feed-in tariff decrease after warranty	0.00 %

Return on investment

Payback period	14.3 years
Net present value (NPV)	350,052,075.97 EUR
Internal rate of return (IRR)	0.00 %
Return on investment (ROI)	107.1 %

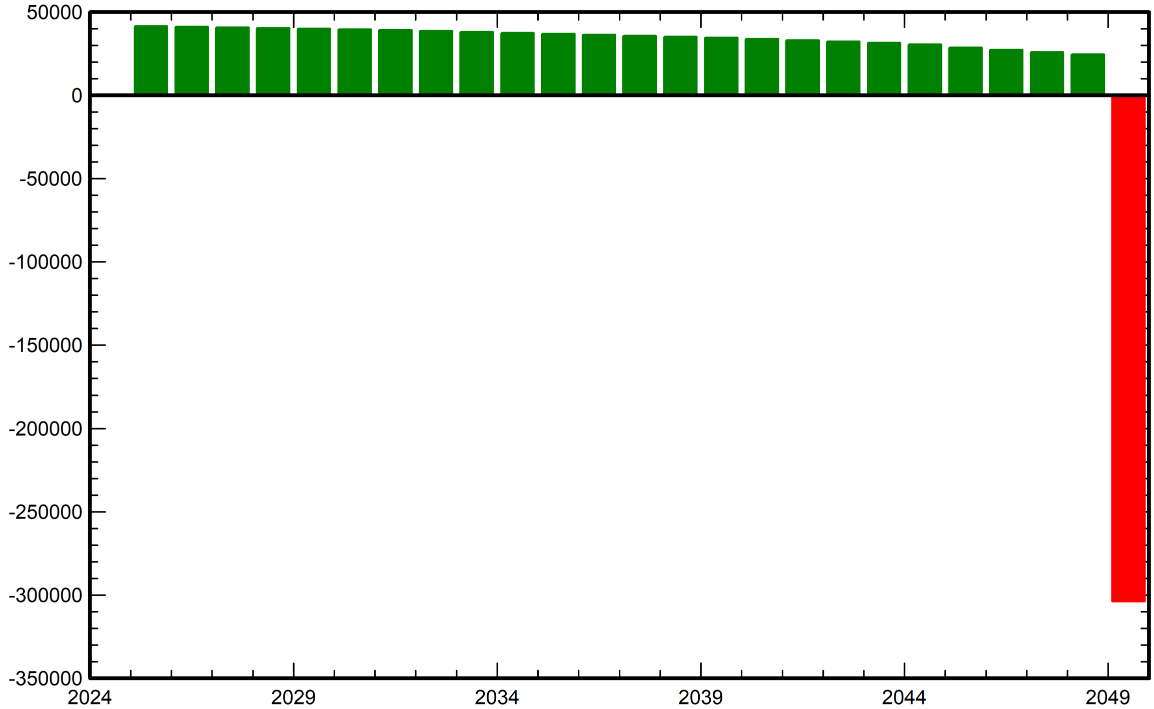
Financial analysis

Detailed economic results (kEUR)

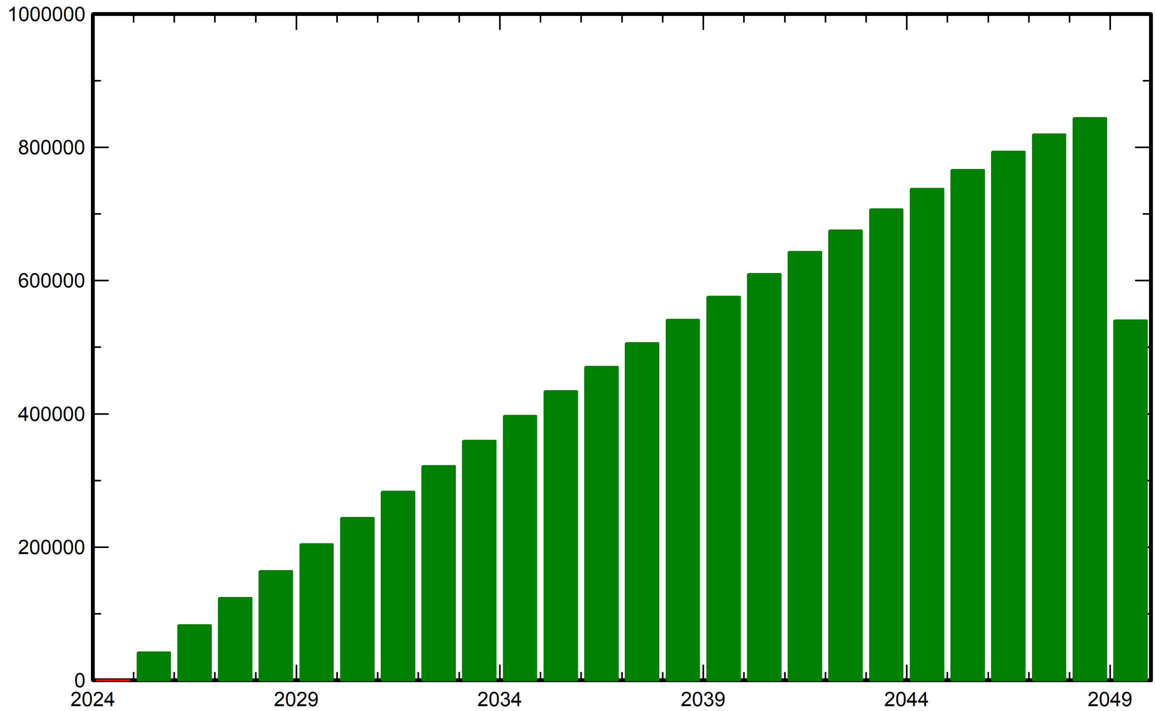
Year	Electricity sale	Loan principal	Loan interest	Run. costs	Deprec. allow.	Taxable income	Taxes	After-tax profit	Cumul. profit	% amorti.
0	0	0 0	0	0	0	0	0	0	0	0.0%
1	85,626,769	0 0	7,141,381	2,153,002	2,254,125	74,078,261	34,816,783	41,515,603	38,440,374	11.8%
2	85,195,526	0 0	7,141,381	2,368,302	2,254,125	73,431,718	34,512,908	41,172,936	73,739,530	22.6%
3	84,764,283	0 0	7,141,381	2,605,132	2,254,125	72,763,645	34,198,913	40,818,857	106,142,854	32.5%
4	84,333,040	0 0	7,141,381	2,865,645	2,254,125	72,071,889	33,873,788	40,452,226	135,876,448	41.6%
5	83,901,798	0 0	7,141,381	3,152,210	2,254,125	71,354,082	33,536,418	40,071,788	163,148,634	49.9%
6	83,470,555	0 0	7,141,381	3,467,431	2,254,125	70,607,618	33,185,581	39,676,163	188,151,347	57.6%
7	82,840,740	0 0	7,141,381	3,814,174	2,254,125	69,631,060	32,726,598	39,158,587	211,000,006	64.6%
8	82,210,925	0 0	7,141,381	4,195,591	2,254,125	68,619,828	32,251,319	38,622,634	231,866,613	70.9%
9	81,581,110	0 0	7,141,381	4,615,150	2,254,125	67,570,453	31,758,113	38,066,465	250,909,323	76.8%
10	80,951,295	0 0	7,141,381	5,076,665	2,254,125	66,479,123	31,245,188	37,488,060	268,273,548	82.1%
11	80,321,479	0 0	7,141,381	5,584,332	2,254,125	65,341,642	30,710,572	36,885,195	284,092,976	86.9%
12	79,866,167	0 0	7,141,381	6,142,765	2,254,125	64,327,896	30,234,111	36,347,910	298,527,232	91.3%
13	79,410,855	0 0	7,141,381	6,757,041	2,254,125	63,258,308	29,731,405	35,781,028	311,683,841	95.4%
14	78,955,543	0 0	7,141,381	7,432,745	2,254,125	62,127,291	29,199,827	35,181,589	323,661,802	99.0%
15	78,500,231	0 0	7,141,381	8,176,020	2,254,125	60,928,705	28,636,491	34,546,339	334,552,249	102.4%
16	78,044,919	0 0	7,141,381	8,993,622	2,254,125	59,655,791	28,038,222	33,871,694	344,439,073	105.4%
17	77,573,560	0 0	7,141,381	9,892,984	2,254,125	58,285,070	27,393,983	33,145,212	353,397,195	108.1%
18	77,102,202	0 0	7,141,381	10,882,282	2,254,125	56,824,413	26,707,474	32,371,064	361,498,022	110.6%
19	76,630,843	0 0	7,141,381	11,970,511	2,254,125	55,264,827	25,974,468	31,544,483	368,807,260	112.8%
20	76,159,485	0 0	7,141,381	13,167,562	2,254,125	53,596,417	25,190,316	30,660,226	375,385,356	114.9%
21	75,688,127	0 0	7,141,381	14,484,318	0	54,062,427	25,409,341	28,653,087	381,077,456	116.6%
22	74,712,816	0 0	7,141,381	15,932,750	0	51,638,685	24,270,182	27,368,503	386,111,633	118.1%
23	73,737,505	0 0	7,141,381	17,526,025	0	49,070,099	23,062,947	26,007,153	390,541,048	119.5%
24	72,762,194	0 0	7,141,381	19,278,627	0	46,342,186	21,780,827	24,561,359	394,414,358	120.7%
25	71,786,884	0 326,836,667	7,141,381	21,206,490	0	43,439,013	20,416,336	-303,813,991	350,052,076	207.1%
Total	**1,986,128,849**	**326,836,667**	**178,534,529**	**211,741,372**	**45,082,500**	**1,550,770,447**	**728,862,110**	**540,154,170**	**350,052,076**	**207.1%**

Financial analysis

Yearly net profit (kEUR)



Cumulative cashflow (kEUR)



CO_2 Emission Balance

Total: 1259732.2 tCO_2

Generated emissions

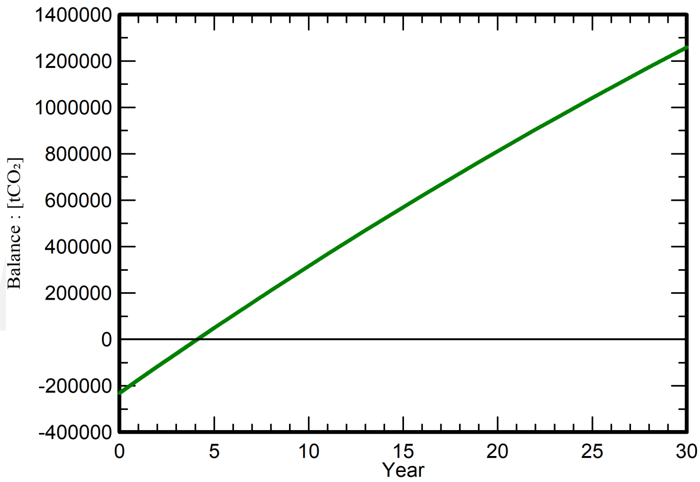
Total: 231648.68 tCO_2

Source: Detailed calculation from table below

Replaced Emissions

Total: 1718843.2 tCO_2
System production: 199633.36 MWh/yr
Grid Lifecycle Emissions: 287 gCO_2/kWh
Source: IEA List
Country: Spain
Lifetime: 30 years
Annual degradation: 1.0 %

Saved CO_2 Emission vs. Time



System Lifecycle Emissions Details

Item	LCE	Quantity	Subtotal
			[$kgCO_2$]
Modules	1713 kgCO2/kWp	132693 kWp	227265921
Supports	1.91 kgCO2/kg	2287810 kg	4379532
Inverters	190 kgCO2/	17.0	3224